

Maximizing/Minimizing Submodular Fns

- We have seen: maximizing submodular f ; monotone function subject to matroid constraint:
- ① hard - 3 examples where approx exact solution (or even arb. approx) is NP-hard.
 - ② Greedy gives $(1 - 1/e)$ -approx for max s.t. unif matroid - best possible.
 - ③ Greedy gives $\frac{1}{2}$ -approx for max s.t. matroid constraint.
- Continuous extensions map a different approach for max/min.

Continuous Extensions

$$f: 2^E \rightarrow \mathbb{R}, \text{ equiv: } f: \{0,1\}^{|E|} \rightarrow \mathbb{R}$$

we ask: how can we extend to a function $[0,1]^{|E|} \rightarrow \mathbb{R}$.

Two canonical continuous extensions:

Concave Closure: $f^+(x) \triangleq \max \left\{ \sum_{s \in E} \alpha_s f(s) : \alpha_s \geq 0, \sum_s \alpha_s = 1, \text{ and } \sum \alpha_s \cdot x^s = x \right\}$

Convex Closure: $f^-(x) \triangleq \min \left\{ \sum_{s \in E} \alpha_s f(s) : \alpha_s \geq 0, \sum_s \alpha_s = 1, \text{ and } \sum \alpha_s \cdot x^s = x \right\} \textcircled{*}$

Note: Coeff α_s can be interpreted as a prob dist'n over 2^E .

Concave and Convex Extensions

Lemma: f^+ is concave.

$$x_1, x_2 \in [0, 1]^{|E|} : f^+(x_1) = \sum \beta_s f(s)$$

$$f^+(x_2) = \sum \gamma_s f(s)$$

$$\begin{aligned} \underline{x} = \lambda \underline{x}_1 + (1-\lambda) \underline{x}_2 & \xrightarrow{\quad} \sum (\lambda \beta_s + (1-\lambda) \gamma_s) x^s = x \\ f^+(x) = \sum \delta_s f(s) \end{aligned}$$

$$\geq \sum \lambda \beta_s f(s) + (1-\lambda) \gamma_s f(s)$$

$$= \lambda f^+(x_1) + (1-\lambda) f^+(x_2).$$

$\Rightarrow f^+$ is concave.

Lemma: f^- is convex.

Lovász Extension — 1/2

For any $x \in [0,1]^{|E|}$

re-order the coefficients so that they are in increasing order:

$$0 \leq x_1 \leq x_2 \leq \dots \leq x_{|E|}$$

S_1 = indices of largest value

$S_2 = S_1 \cup$ indices of next largest value

etc.

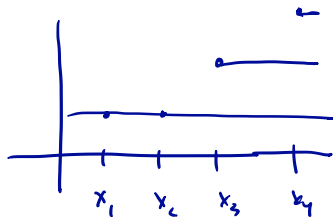
Example: $x = \begin{pmatrix} 0 \\ 1/4 \\ 1/4 \\ 1/3 \\ 1/2 \\ 1/2 \end{pmatrix}$

$$S_1 = \{5, 6\}$$

$$S_2 = \{4, 5, 6\}$$

$$S_3 = \{2, 3, 4, 5, 6\}$$

$$S_4 = \{1, 2, 3, 4, 5, 6\}$$



$$\begin{aligned} //x &= \left(\frac{1}{2} - \frac{1}{3}\right) \cdot x^{S_1} + \left(\frac{1}{3} - \frac{1}{4}\right) \cdot x^{S_2} + \left(\frac{1}{4} - 0\right) \cdot x^{S_3} + 0 \cdot x^{S_4} \\ &= \left(\frac{1}{2} - \frac{1}{3}\right) \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 1 \\ 1 \end{pmatrix} + \left(\frac{1}{3} - \frac{1}{4}\right) \begin{pmatrix} 0 \\ 0 \\ 1 \\ 1 \\ 1 \\ 1 \end{pmatrix} + \left(\frac{1}{4} - 0\right) \begin{pmatrix} 0 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{pmatrix} + 0 \cdot \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{pmatrix} \end{aligned}$$

Ex: We can always write x in this way.

Lovász Extension — 2/2

The prev. claim, says: given $f: \{0,1\}^6 \rightarrow \mathbb{R}$
we can now extend to $[0,1]^6$ by applying f linearly to RHTs.

$$f^L(x) = \underbrace{(\frac{1}{2} - \frac{1}{3})}_{\lambda_1} f(S_1) + \underbrace{(\frac{1}{3} - \frac{1}{4})}_{\lambda_2} f(S_2) + \underbrace{(\frac{1}{4} - 0)}_{\lambda_3} f(S_3)$$

Def'n: Lovász Extension

Given $f: \{0,1\}^E \rightarrow \mathbb{R}$

$$\forall x \in [0,1]^E, f^L(x) = \sum \lambda_i f(S_i)$$

where $\lambda_i \geq 0$, $\sum \lambda_i = 1$, $\emptyset = S_0 \subseteq S_1 \subseteq S_2 \subseteq \dots \subseteq S_n$

Note: f^L is easy to evaluate.

① $x \rightarrow$ find S_i, λ_i

② Evaluate $f(S_0), f(S_1), \dots, f(S_n)$.

Lovász Extension and Submodularity

Thm: $f^L(x) = f^-(x)$ iff f is submodular.

pf: " $\Leftarrow$ " Suppose f is submodular.

Let's take $\{\alpha_S\}$ to be the distribution achieving $f^-(x)$,

that also maximizes: $\sum \alpha_S \cdot |S|^2$. //

Claim: This $\{\alpha_S\}$ is supported on a chain. If not, then $A, B \in E$

$\alpha_A \geq \alpha_B > 0$, $A \not\subseteq B$, $B \not\subseteq A$. By submodularity: $f(A \cup B) + f(A \cap B) \leq f(A) + f(B)$.

Replace α_B of A and B by α_B of $A \cup B$ and $A \cap B$: i.e., define $\{\gamma_S\}$,
 $\gamma_S = \alpha_S$ except: $\gamma_B = 0$, $\gamma_A = \alpha_A - \alpha_B$, $\gamma_{A \cup B} = \alpha_{A \cup B} + \alpha_B$, $\gamma_{A \cap B} = \alpha_{A \cap B} + \alpha_B$.

$\sum \alpha_S f(S) \geq \sum \gamma_S f(S)$, but: $\sum \alpha_S |S|^2 < \sum \gamma_S |S|^2$ // Exercise.

Contradiction shows $\text{supp}(\alpha_S)$ is nested.

But: The decomp into a chain is unique $\Rightarrow f^-(x) = f^L(x)$.

" $\Rightarrow$ " Suppose f is not submodular.

Hence: $\exists S, i, j \in E \setminus S$ where:

$$f(S) + f(S+i+j) > \underline{f(S+i) + f(S+j)} //$$

$$x \in [0,1]^{|E|}, \quad x = x^S + \frac{1}{2} x^{\{i,j\}} = \begin{pmatrix} 1 \\ \vdots \\ \frac{1}{2} \\ \vdots \\ 0 \end{pmatrix} \quad \begin{matrix} | \\ | \\ | \\ | \\ | \end{matrix}$$

$$f^L(x) = \frac{1}{2} f(S) + \frac{1}{2} f(S \cup \{i,j\})$$

$$S_1 = S \cup \{i\}, \quad S_2 = S \cup \{j\},$$

$$\alpha_{S_1} = \gamma_L = \alpha_{S_2}.$$

$$f^-(x) \leq \frac{1}{2} f(S+i) + \frac{1}{2} f(S+j) < f^L(x).$$



Minimizing Submodular Functions